

Suppose you have a coin that your priors say is probably fair. You then flip the coin twice and observe: heads, heads

Without doing any calculations,

This is what I was thinking of in the previous lesson where you said my intuition was backwards comparing MAP and MLE. I was answering about my credences. I was answering the wrong question.

1) Compared to before the flips, should the posterior shift towards larger values of θ ?

Slightly, because in worlds where the coin is fair, seeing two heads in a row is expected. Especially with the additional prior that unfair coins are relatively rare. The data is much more easily attributed to noise than a real signal. It's very weak evidence of an unfair coin.

2) Should your confidence be high or low after only two flips? Why?

By this point I'd still be pretty confident of my priors. Two flips would be weak evidence for an unfair coin, and I would

Not put much credence in the hypothesis that something is wrong with the coin.

- 3) If you instead flip the coin 2000 times with about 60% landing on heads, what would happen to the influence of the prior?

By 2000 flips, I would be pretty confident that the coin is not fair. I still wouldn't rule out the possibility that the extra 200 flips over my expectation of 1000 heads was due to experimental noise, rather than an unfair coin, but I'd be pretty confident that it is a fraud. In other words, my posterior would shift towards the data.

- 4) In your own words, why would a Bayesian prefer to keep the whole posterior distribution instead of reporting a single number?

Repeated samples would narrow down hypotheses, but the space of hypotheses is generally large. For instance, as in the previous example the coin may be fair, but the person flipping it may be using a throwing technique that biases the outcome. The coin could be weighted, or have some weird edge shape that could influence the

outcome. Perhaps Tymora is interceding, affecting how the coin lands,

Each of these hypotheses will have a different credence attached to them. Tymora is entirely fictional, so that hypothesis is very unlikely. Weight-shifting a coin so that it is biased in a way that isn't noticeable would be possible, but fairly rare. I think it is quite possible that someone could practice a technique to influence the outcome of the coin toss 10% of the time.

I couldn't quite rule out that I'm trapped in a simulated world run by sadistic statisticians, but if all coins started behaving in this way, I'd give it more credence.